

UNM-PNM Round 1

Solutions

Sean Choi

November 2024

Solution 1. Note that $\frac{2023}{2024} = 1 - \frac{1}{2024}$ and that $\frac{2024}{2025} = 1 - \frac{1}{2025}$. Since $\frac{1}{2024} > \frac{1}{2025}$, we have

$$\frac{2023}{2024} = 1 - \frac{1}{2024} < 1 - \frac{1}{2025} = \frac{2024}{2025}.$$

Thus, our answer is 2025.

Solution 2. By inspection note that 2 is the only valid units digit of the smaller number. The digits 4, 5, 7 will not work since the leftmost digit of the multiple will not be one of 2, 4, 5, or 7 or the multiple will be a 5 digit number.

Now, since the largest possible number is 7542 and the smallest possible number is 2457, we only need to consider multiplication by 2 or 3. Otherwise the multiple will clearly exceed 7542.

Next, we consider the last digit. If we are multiplying by 2, the last digit can go from 5 to 0, 7 to 4, or from 4 to 8. From these possibilities, only going to 7 to 4 satisfies the conditions given in the problem. However, it is easy to check that neither 2457 nor 2547 work.

When we multiply by 3, the units digit can go from 5 to 5, 7 to 1, or from 4 to 2. The only cases that work are when the units digit goes from 5 to 5 and from 4 to 2. Then, we can check the possibilities:

$$2475, 2745, 2574, 2754.$$

From these, note that $2475 \cdot 3 = 7425$, which is what we are looking for. Thus, our answer is $2475 + 7425 =$ 9900.

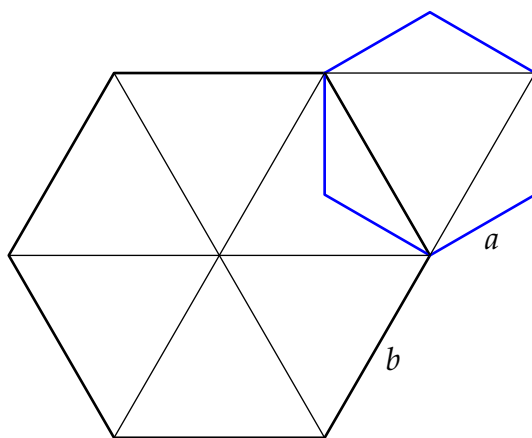
Solution 3. The optimal strategy for Cora is to force the game from a 3-pile game to a 2-pile game. This means that Cora should take all the stones from the first pile.

Then, if Ernie is playing optimally as well, Ernie must take some n stones ($n < 7$) from one of the two piles remaining. Otherwise (i.e., if $n = 7$), the game will become a one-pile game from which Cora will be able to win by taking all the stones.

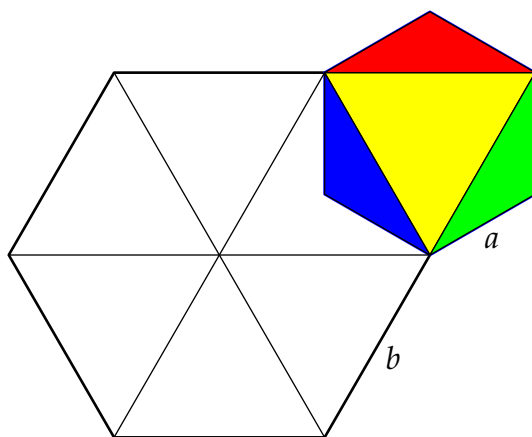
Then, Cora can take n but from the other pile, effectively "mirroring" Ernie's moves. If Cora plays in this manner, Cora will be able to reduce the game to a situation where the second player has two single-stone piles left. This means that they must take one stone from a pile, leaving Cora to win the game by taking the last stone.

So, Cora will always be able to win when playing optimally. Thus, the answer is 1.

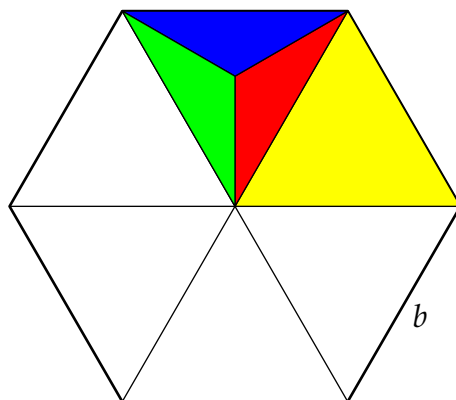
Solution 4. A visual solution. First, we decompose the hexagons as follows:



In order to clearly see what is going on, we will add a bit of color:



Rearranging the areas, we have:

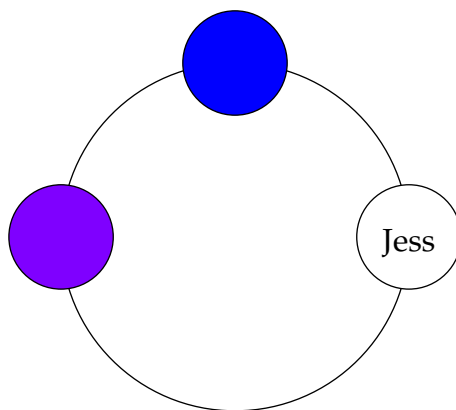


Thus, the ratio of the areas is $\frac{6}{2} = \boxed{3}$.

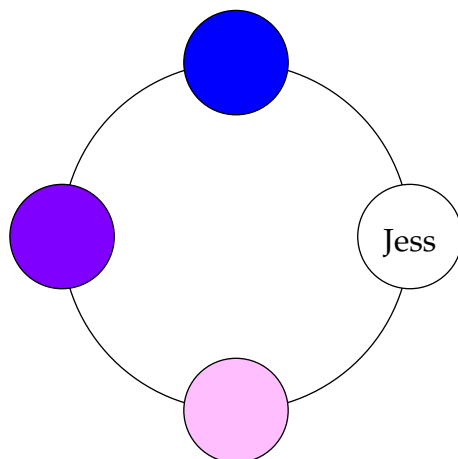
Solution 5. Note that $x = \sqrt{1 + \sqrt{1 + \sqrt{1}}} = \sqrt{1 + \sqrt{2}}$. Now,

$$\begin{aligned} x^4 - 2\sqrt{2} &= (x^2)^2 - 2\sqrt{2} = (1 + \sqrt{2})^2 - 2\sqrt{2} \\ &= (1 + 2\sqrt{2} + 2) - 2\sqrt{2} \\ &= \boxed{3}. \end{aligned}$$

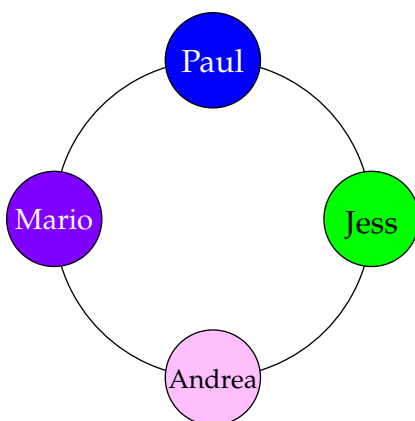
Solution 6. Given the first condition, we can let the person with purple hair, the person with blue hair, and Jess be 3 people in order on the circle:



Now, we consider who can have pink hair. If Jess has Pink hair, then Mario must have blue hair, a contradiction. Thus, the person with pink hair must take the last remaining spot in the circle:



This means that Jess must have green hair and that Mario has purple hair. Since Andrea cannot have blue hair, Paul must have blue hair and Andrea must have pink hair:



Filling in the table according to the numbers corresponding to the hair colors given, we find that the answer is 4321.

Solution 7. Let

a = lifespan of bowhead whale

b = lifespan of brown bear

c = lifespan of western gorilla

d = lifespan of wolverine

e = lifespan of wild turkey

f = lifespan of tiger

g = lifespan of elephant.

From the information given in the problem, we construct the system of equations

$$\begin{cases} a &= 8b \\ b &= c - 11 \\ c &= 3d \\ d &= e + 8 \\ e &= f - 10 \\ f &= \frac{1}{4}g \\ g &= \frac{a + b + c + d + e + f + g - 11}{6}. \end{cases}$$

Starting from the fourth equation, we can work upwards and write a, b, c , and d in terms of e . We can also write the 5th and 6th equations in terms of e , which is what we wish to find. Doing so yields

$$\begin{cases} a &= 24e + 104 \\ b &= 3e + 13 \\ c &= 3e + 24 \\ d &= e + 8 \\ f &= e + 10 \\ g &= 4e + 40. \end{cases}$$

Then, we can substitute into

$$g = \frac{a + b + c + d + e + f + g - 11}{6}$$

and simplify to get

$$4e + 40 = \frac{1}{6}(37e + 188).$$

This gives

$$24e + 240 = 37e + 188$$

which yields the solution $e = \boxed{4}$ for the lifespan of a wild turkey.

Solution 8. Since n is nonnegative,

$$n^2 - 11n + 25 < n^2 - 10n + 25 = (n - 5)^2.$$

Now, note that the closest perfect square below $(n - 5)^2$ is

$$(n - 6)^2 = ((n - 5) - 1)^2 = (n - 5)^2 - 2(n - 5) + 1.$$

Then, if

$$(n - 6)^2 < n^2 - 11n + 25 < (n - 5)^2$$

is satisfied, then $n^2 - 11n + 25$ will always be between two consecutive perfect squares, which means that it can never be a perfect square. Simplifying the inequality, we get

$$\begin{aligned} n^2 - 12n + 36 &< n^2 - 11n + 25 < n^2 - 10n + 25 \\ -12n + 36 &< -11n + 25 < -10n + 25 \end{aligned}$$

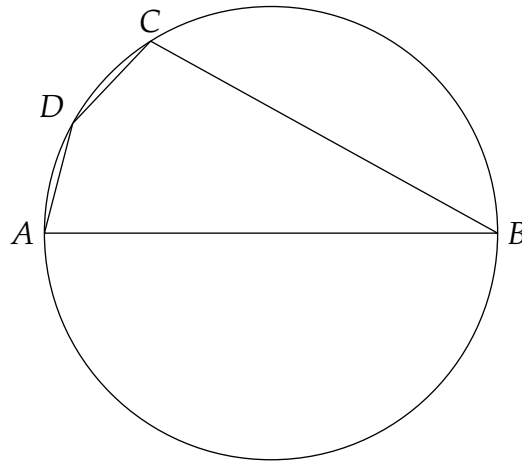
So $n > 11$ is when this is satisfied. Thus, it suffices to test $n \leq 11$.

Now, we take the expression modulo 4 in order to narrow down the cases we test. We have

$$n^2 - 11n + 25 \equiv n^2 + n + 1 \pmod{4}.$$

Testing residues modulo 4, the only two residues that work are $n \equiv 0 \pmod{4}$ and $n \equiv 3 \pmod{4}$, which means that we only have to test $n = 0, 4, 8, 3, 7, 11$. We find that $n = 4, n = 7$ do not yield perfect squares so our answer is $0 + 8 + 3 + 11 = \boxed{22}$.

Solution 9. Let the radius be r . Then $AB = 2r$. Since $\overline{AB}$ is the diameter of the circle, $\triangle ABD$ and $\triangle ABC$ are right. Thus, applying the Pythagorean Theorem, we have that $AC = \sqrt{(2r)^2 - 14^2}$ and $BD = \sqrt{(2r)^2 - 4^2}$.



Since $ABCD$ is a cyclic quadrilateral, we can apply Ptolemy's theorem, which states that $AC \cdot BD = AB \cdot CD + BC \cdot AD$. So,

$$\sqrt{(2r)^2 - 14^2} \cdot \sqrt{(2r)^2 - 4^2} = (2r) \cdot 4 + 14 \cdot 4.$$

Simplifying, we have

$$\sqrt{(2r)^2 - 14^2} \cdot \sqrt{(2r)^2 - 4^2} = 8r + 56,$$

$$\sqrt{r^2 - 49} \cdot \sqrt{r^2 - 4} = 2r + 14.$$

Squaring both sides,

$$(r^2 - 49)(r^2 - 4) = (2r + 14)^2,$$

$$(r + 7)(r - 7)(r^2 - 4) = 4(r + 7)^2,$$

$$(r + 7)(r^3 - 7r^2 - 4r + 28) = (r + 7)(4r + 28),$$

$$(r + 7)(r^3 - 7r^2 - 8r) = 0,$$

$$r(r + 7)(r^2 - 7r - 8) = 0.$$

Factoring,

$$r(r + 7)(r - 8)(r + 1) = 0.$$

We take the positive solution, so $r = \boxed{8}$.

Solution 10. We have

$$u^2 = x^2 \left(\frac{1 + r^{-2}}{2} \right)^2$$

and

$$v^2 = y^2 \left(\frac{1 - r^{-2}}{2} \right)^2.$$

Then,

$$x^2 = u^2 \left(\frac{2}{1+r^{-2}} \right)^2$$

and

$$y^2 = v^2 \left(\frac{2}{1-r^{-2}} \right)^2.$$

Then, we map the equation for the circle

$$\frac{x^2}{r^2} + \frac{y^2}{r^2} = 1$$

to

$$\frac{u^2 \left(\frac{2}{1+r^{-2}} \right)^2}{r^2} + \frac{v^2 \left(\frac{2}{1-r^{-2}} \right)^2}{r^2} = 1.$$

Then, note that

$$\left(\frac{2}{1+r^{-2}} \right)^2 \cdot \frac{1}{r^2} = \frac{1}{a^2}$$

and

$$\left(\frac{2}{1-r^{-2}} \right)^2 \cdot \frac{1}{r^2} = \frac{1}{b^2}.$$

Thus, since $a = \frac{5}{3}$ and $b = \frac{4}{3}$, we have

$$\left| \frac{2}{r+r^{-1}} \right| = \frac{3}{5}$$

and

$$\left| \frac{2}{r-r^{-1}} \right| = \frac{3}{4}.$$

Then,

$$r + r^{-1} = \frac{10}{3}$$

and

$$|r - r^{-1}| = \frac{8}{3}.$$

From the first equation, we get $r = \frac{1}{3}$ and $r = 3$. We can confirm that both these values satisfy the second equation as well. Thus, the desired sum is $\frac{10}{3}$ and so the answer is 10.